

Data-Driven Forecasting and Control of Greenhouse Microclimate: From Operating-Regime Discovery to Deployment-Aware Multi-Horizon Prediction

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1 Introduction

Protected cultivation gives growers direct control over the aerial environment a crop experiences, but that control is only as good as the information the grower – or an automated controller – has about how the greenhouse will respond to it. Interior temperature and relative humidity jointly determine plant transpiration and disease pressure through the vapour pressure deficit (VPD) they induce (Yu et al., 2023), while the climatic systems available to influence them – heating, forced-air fog, roof and side vents, shade screens, recirculators – interact with exterior weather, time of day and each other in ways that are difficult to specify from first principles alone (Ortiz et al., 2023). This gap between what a grower needs to know and what first-principles models can cheaply provide is what motivates a data-driven alternative: physically-grounded energy-balance models encode the causal structure of ventilation and radiation exchange but require site-specific calibration effort that rarely scales across greenhouses, whereas models learned directly from historical sensor logs avoid that calibration cost at the price of inheriting whatever biases are present in how the greenhouse was actually operated. This paper takes the data-driven route, and is explicit, throughout, about where that route’s evidence stops being causal.

Sequence models – long short-term memory networks (LSTM) (Hochreiter and Schmidhuber, 1997), gated recurrent units (GRU) (Cho et al., 2014), and attention-based architectures (Vaswani et al., 2017) – are by now standard tools for multi-step time series forecasting and have already been applied to greenhouse climate prediction alongside classical tabular learners such as random forests (Breiman, 2001) and gradient-boosted trees (Ke et al., 2017). A recurring methodological difficulty in this literature, however, is that architecture comparisons are frequently run with mismatched target scaling, mismatched validation criteria, a single random seed, or without accounting for how large a model actually is – any of which can make one architecture look better or worse than it really is, a concern that recent large-scale forecasting benchmarks have shown to be widespread rather than anecdotal (Qiu et al., 2024). Separately, reinforcement learning has recently been explored for greenhouse climate control (Mansour et al., 2025), almost always in simulation, since letting an untrained agent act on a live crop is not an option; the fidelity of that simulator is then the single factor that determines whether the learned policy means anything at all. Both issues – a fair architecture comparison, and an honest accounting of what a simulator-based controller can and cannot claim – are treated together in this paper rather than in isolation, because they compound: an architecture ranking that is already unreliable becomes actively misleading once it is used to justify which model backs a downstream controller.

This paper works through both issues on a single, shared dataset – one growing season of hourly data from a research greenhouse recording interior climate at three heights, thirteen actuator channels, an on-site exterior weather station, and a numerical weather forecast product – organised as four linked analyses:

- (i) **Operating-regime discovery.** We cluster each hour’s snapshot (interior climate, actuator activity, exterior weather, time of day) into a small number of interpretable operating states – *cold night*, *heating on*; *hot midday*, *ventilating/shading*; *humidifying*, and so on – fitting the standardisation and the clustering itself on the training period only, so the resulting label carries no information from the future and can safely be used as a forecasting feature.
- (ii) **Multi-horizon, multi-output climate forecasting.** We predict interior temperature and humidity at +1 h, +6 h and +12 h simultaneously, comparing tabular and neural architectures under a feature-ablation ladder (actuators, actuator history, operating regime, day-ahead forecast) *and* under an explicit deployment-information-regime axis (persistence / snapshot / historical / day-ahead forecast / oracle) that separates the value of more history or a better architecture from the value of simply being allowed to know more. This is the paper’s methodological core and is described in full in Section 2.
- (iii) **What-if sensitivity analysis.** Using the forecasting models as a queryable surrogate, we sweep each actuator’s setting while holding the rest of the state fixed and read off the model’s predicted effect – a partial-dependence style analysis – and directly measure how much this analysis is compromised by the fact that several actuators were run *in response* to the very conditions they are assumed to regulate.
- (iv) **A reinforcement-learning control demonstration.** We build a one-step simulator from the +1 h forecasting models, define a compact macro-action space and a setpoint-tracking reward, and train a tabular Q-learning agent (Watkins and Dayan, 1992; Sutton and Barto, 2018), evaluating it against passive and greedy model-predictive-control baselines. We treat this explicitly as a demonstration of the RL machinery rather than a deployable controller, for the reason established in (iii).

Throughout, we prioritise methodological honesty about what observational data can and cannot support over headline accuracy numbers: Section 2 details the leakage-safety guarantees, the fairness corrections applied to the architecture comparison, and the confounding diagnostics behind (iii)–(iv), so that every later result can be read against the precise conditions under which it was obtained.

2 Materials and Methods

2.1 Dataset and preprocessing

The dataset merges, onto a single regular hourly index, every interior sensor channel (temperature, relative humidity, pressure, radiation and wind at high/middle/low positions inside the greenhouse), thirteen actuator channels reported as minutes-run-per-hour (heaters, forced-air fog, roof and side vents, shade screens, recirculators, blowers), an on-site exterior weather station, and an external numerical weather forecast product. Missing actuator readings are treated as *off* (the physically correct default); short gaps in continuous sensor channels are forward-filled from *past* observations only, so that the resulting series remains valid for online, deployable use (no value is ever filled in from the future). The external forecast product provides two kinds of columns: an un-suffixed “latest run” estimate that turns out to be closer to a same-time reanalysis than to a genuine advance forecast, and a `previous_day1` column issued with a real

$\sim$ 1-day lead time; only the latter is used as a forecasting feature (Section 2.3), and any residual gaps in it are forward-filled with an explicit *stale* flag rather than backfilled from the same-time estimate, which would silently break the forecast’s lead-time guarantee. The whole study is scoped to the period from which every climatic system is actively logged (`ACTIVE_START`), so that actuators are not spuriously "unhelpful" simply because their channel was all-zero before installation/instrumentation was complete.

All analyses share a single *chronological* train/test split (no shuffling of a time series), computed once and reused across notebooks so that no component – clustering, forecasting, or the control studies – can leak information across that boundary into another.

2.2 Operating-regime clustering

Each hour is described by four feature families: interior climate, actuator activity, exterior weather, and a cyclic (sine/cosine) encoding of time-of-day, deliberately excluding the future temperature/humidity targets. Features are z -scored and clustered with k -means; critically, *both* the scaler and the k -means model are fit on the training period only and only ever *applied* (never refit) to the test period, using the same chronological boundary as the forecasting split (Section 2.1) – otherwise the cluster centroids would themselves encode test-period information, a subtler form of the same leakage the forecasting models must avoid. The number of clusters was chosen by balancing the elbow/silhouette diagnostics (computed on the training period only) against interpretability for control purposes, favouring *granularity* over the raw silhouette optimum; this yields $k = \text{[TODO: 8]}$ operating states (e.g. *cold night*, *heating on*; *hot midday*, *ventilating/shading*; *humidifying*), each characterised by its mean temperature/humidity and actuator pattern. The resulting per-hour label is saved and reused in Section 2.3 as a *snapshot*-regime input feature (Section 2.3.5): because it is computed from information available at the same hour only, it is safe to feed to a forecaster without introducing leakage.

2.3 Multi-horizon, multi-output climate forecasting

2.3.1 Problem framing and feature ablation ladder

The core prediction task is a multi-output regression: interior temperature and humidity at +1h, +6h and +12h (six targets jointly). Models are compared along an escalating *feature-ablation ladder* – *no actuators* $\rightarrow$ *+ current actuators* $\rightarrow$ *+ actuator history* (rolling 3h/6h sums, capturing thermal/humidity inertia that a single on/off reading cannot) $\rightarrow$ *+ operating-regime label* (Section 2.2) $\rightarrow$ *+ day-ahead forecast* – so that the marginal value of each information source is measured rather than assumed.

2.3.2 Architecture grid

Two families of learners are trained on every rung of the ladder. The *tabular* family (ridge regression, random forest (Breiman, 2001), and histogram-based gradient boosting (Ke et al., 2017)) is given either a single row of "now" (*t only*) or the same 24-hour window flattened into one row (*24h hist*), so that its access to history can be compared on equal footing with the sequence models. The *neural* family – an LSTM (Hochreiter and Schmidhuber, 1997), a GRU (Cho et al., 2014), a compact Transformer-style multi-head self-attention block (Vaswani et al., 2017) (deliberately scaled down from a full Temporal Fusion Transformer, to keep the parameter count in the same range as the other sequence models rather than adding machinery whose main effect on $\sim$ 1,200 training rows would be to overfit), and a plain feed-forward (flatten + dense) baseline with no notion of temporal order at all – is likewise trained with a 24-hour lookback and, separately, with `lookback=1`, completing a $\{\text{tabular, LSTM, GRU, attention, MLP}\} \times \{t \text{ only, 24h history}\}$ grid. All sequence inputs are built by the same windowing routine, with the feature/target scaler fit on training rows only and applied (never refit) to the full series, so that a test-period sequence’s

window may reach back into the training period but no training computation ever sees a test value.

2.3.3 Hyperparameter tuning and model selection

Tabular models search a small fixed grid (ridge’s α ; random forest’s minimum leaf size; boosting’s learning rate) on a chronological hold-out carved from the end of the training period, never touching the test set. Every neural architecture is tuned identically via Bayesian optimisation (Snoek et al., 2012) over its width, dropout rate, learning rate and optimiser, using the same validation split, early-stopping criterion and trial budget for all four architectures, so that no architecture is searched more generously than another. Both hyperparameter selection and every reported "best model" ranking use the *mean CVRMSE* (root-mean-squared error normalised by the target’s mean, unitless) across the six targets, rather than raw mean absolute error (MAE): temperature error ($^{\circ}\text{C}$) and humidity error ($\%\text{RH}$) are different physical quantities and are never averaged together to make a selection decision. MAE, CVRMSE and the true coefficient of determination R^2 (Pedregosa et al., 2011) (not squared Pearson correlation, which is invariant to affine bias and can therefore overstate agreement) are reported per target throughout.

2.3.4 Fair architecture comparison across seeds and parameter budgets

Because the four neural architectures share the same hyperparameter search range (e.g. hidden width) but do *not* thereby have comparable parameter counts – the feed-forward baseline’s input layer scales with $\text{lookback} \times n$ features, while the recurrent layers’ size does not – every architecture’s parameter count is reported alongside its accuracy. In addition, each architecture’s already-tuned winning configuration is refit from scratch under **[TODO: 5]** different random seeds (no re-tuning, to keep the additional compute bounded), and the resulting mean $\pm$ standard deviation is reported rather than a single run, so that an apparent difference between architectures can be checked against run-to-run variability before it is interpreted as an architecture effect.

2.3.5 Deployment information regimes

The feature-ablation ladder above answers "does this group of columns help a fixed model?", but conflates *how much information a model is allowed at prediction time* with *which architecture is used*. We therefore add an explicit, orthogonal comparison across five *information regimes*: *persistence* (naive carry-forward baseline); *snapshot* (information available at time t only – current climate, actuators and the operating-regime label, no history, no forecast, evaluated with `lookback=1` architectures only, since a 24-hour window would silently reintroduce history); *historical* (snapshot + observed history up to t); *NWP* (historical + the day-ahead forecast, i.e. information a deployed system could genuinely have); and *oracle* (historical + the *actual* future exterior weather in place of the forecast – not deployable, but an upper bound that isolates how much remaining error is attributable to forecast uncertainty rather than to the model itself, by comparison with NWP). Every architecture in the grid above is trained for the snapshot and oracle regimes; historical and NWP reuse the corresponding feature-ablation-ladder predictions (no retraining) but are rescored on the row set this comparison uses, so all five regimes are evaluated on identical rows. A handful of rows lacking a genuine future exterior-weather observation (residual gaps exceeding the short-gap fill of Section 2.1) are excluded consistently from all five regimes rather than only from *oracle*, to keep the comparison on equal footing.

2.3.6 Vapour pressure deficit: regression and critical-event detection

Beyond the two raw sensor targets, we derive the vapour pressure deficit (VPD) from the predicted temperature and humidity via the Tetens equation and score it both as a continuous regression target (MAE, CVMSE, R^2 against the VPD computed from the true future sensor values) and, because growers act on *events* rather than continuous error, as a binary event-detection problem: "VPD too low" (excess humidity, disease/fungal risk) and "VPD too high" (water stress / stomatal closure) are each defined against a fixed threshold ([**TODO: 0.4**]/[**TODO: 1.6**] kPa, literature-typical placeholder values pending validation against agronomic guidance (Yu et al., 2023), fixed independently of the data split so that no test-set information enters the threshold choice) and scored with precision, recall, F1, false-alarm rate, and area under the precision–recall curve (AUPRC) at each horizon, using the model’s own forecast crossing the same threshold as the deployed alarm rule.

2.4 What-if sensitivity analysis

Using the +1 h/ + 6 h forecasting models of Section 2.3 as a queryable surrogate, we sweep one actuator’s setting across its operating range while holding every other input at its real, observed value, and read off the model’s predicted temperature/humidity response – a partial-dependence-style sensitivity curve. Because this dataset is purely observational (no actuator was ever varied independently of the conditions it responds to), the estimated sensitivity curves are, in the language of causal inference (Pearl, 2009), confounded rather than causal effects: we explicitly test for this by computing each actuator’s correlation with interior temperature, since a system switched on *because* it is hot will appear positively correlated with heat regardless of its true cooling effect, which is exactly the pattern that would make a sensitivity curve come out flat or wrong-signed. As a partial mitigation we additionally compare outcomes only within narrow bands of similar exterior conditions and time-of-day (a coarse form of matching), and finally invert the question with a grid search over actuator settings that ranks candidates by predicted closeness to a target temperature/humidity, framed explicitly as a shortlist for an operator rather than a calibrated control law.

2.5 Reinforcement-learning control demonstration

We construct a one-step simulator by training a dedicated +1 h model (Section 2.3’s architecture and leakage-safety protocol, but with the actuator settings exposed as direct, agent-controlled inputs rather than observed features) and use it as the environment for a tabular Q-learning agent (Watkins and Dayan, 1992; Sutton and Barto, 2018) – the same tabular-value-function family used, at larger scale, by recent deep-RL greenhouse controllers (Mansour et al., 2025), chosen here for its transparency rather than its capacity, since a fully human-readable policy table matters more for this demonstration than squeezing out extra performance with a deeper function approximator. The thirteen actuators are collapsed into six interpretable macro-actions (presets inspired by the operating regimes of Section 2.2) to keep the action space tractable; the state is a discretised temperature band $\times$ humidity band $\times$ coarse time-of-day; and the reward penalises distance from a target setpoint ([**TODO: 24 °C**], [**TODO: 65 %RH**]) with temperature and humidity placed on comparable scales. Exterior weather is replayed from real, consecutive historical hours during both training and evaluation, so the agent faces realistic, uncontrollable exogenous conditions and only decides which systems to run. The learned greedy policy is evaluated on held-out start hours against a passive (all-systems-off) baseline, a naive always-ventilate rule, and a one-step greedy model-predictive-control baseline that exhaustively tries every action in the simulator. We treat this study explicitly as a demonstration of the RL machinery – state, action, reward, learning, evaluation, and a human-readable policy table – rather than as a deployable controller, since the simulator is trained on the same confounded,

observational data flagged in Section 2.4, and would need to be replaced by one calibrated on controlled actuation trials or a physical energy-balance model before its recommendations could be trusted.

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